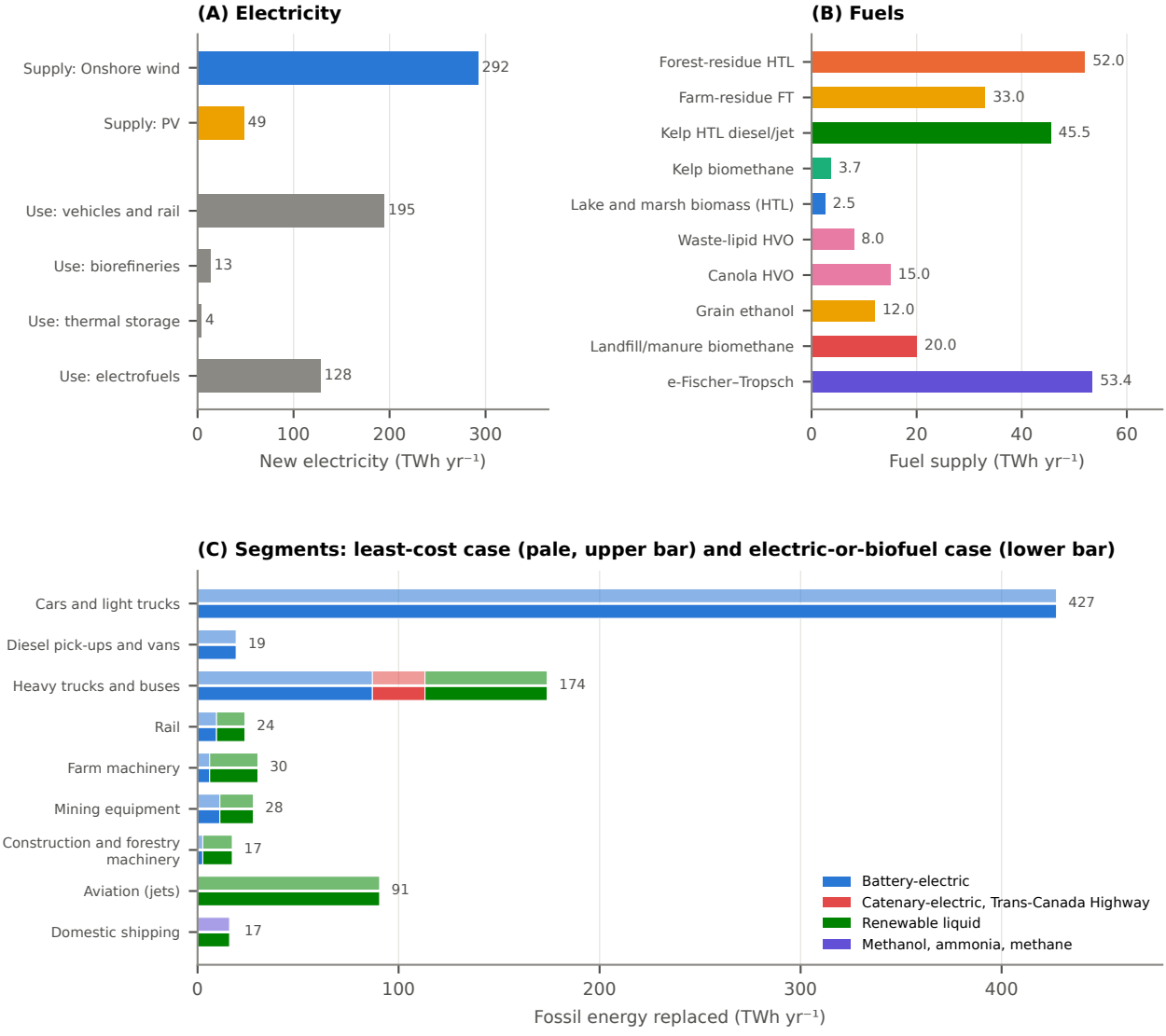
(A) Electricity
Supply: Onshore wind
292
Supply: PV
49
Use: vehicles and rail
195
Use: biorefineries
13
Use: thermal storage
4
Use: electrofuels
128
0
100
200
300
New electricity (TWh yr⁻¹)
(B) Fuels
Forest-residue HTL
52.0
Farm-residue FT
33.0
Kelp HTL diesel/jet
45.5
Kelp biomethane
3.7
Lake and marsh biomass (HTL)
2.5
Waste-lipid HVO
8.0
Canola HVO
15.0
Grain ethanol
12.0
Landfill/manure biomethane
20.0
e-Fischer–Tropsch
53.4
0
20
40
60
Fuel supply (TWh yr⁻¹)
(C) Segments: least-cost case (pale, upper bar) and electric-or-biofuel case (lower bar)
Cars and light trucks
427
Diesel pick-ups and vans
19
Heavy trucks and buses
174
Rail
24
Farm machinery
30
Mining equipment
28
Construction and forestry machinery
17
Aviation (jets)
91
Domestic shipping
17
Battery-electric
Catenary-electric, Trans-Canada Highway
Renewable liquid
Methanol, ammonia, methane
0
100
200
300
400
Fossil energy replaced (TWh yr⁻¹)